



Sydney Girls High School

2026

**Trial Higher School Certificate
Examination**

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using a black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11-14, show relevant mathematical reasoning and/or calculations

**Total marks:
70****Section I – 10 marks**

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 60 marks

- Attempt Questions 11-14
- Allow about 1 hour and 45 minutes for this section

Name:

.....

Teacher:

.....

THIS IS A TRIAL PAPER ONLY

It does not necessarily reflect the format
or the content of the 2026 HSC
Examination Paper in this subject.

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

Questions	Marks
1 What is the magnitude of $2\mathbf{u} + 3\mathbf{v}$?	1
A. $\sqrt{5}$	
B. $\sqrt{11}$	
C. $\sqrt{13}$	
D. Cannot be determined.	
2 The polynomial $P(x) = x^3 + 5x^2 + bx + c$ has a double root at $x = -1$.	1
What are the values of b and c ?	
A. $b = 7, c = 3$	
B. $b = 7, c = -3$	
C. $b = -7, c = 3$	
D. $b = -7, c = -3$	

- 3** What is the value of $\int_0^{\frac{\pi}{4}} (4 \cos^2 x - 5) dx$? **1**
- A. $1 - \frac{3\pi}{4}$
- B. $1 - \frac{5\pi}{4}$
- C. $-1 - \frac{3\pi}{4}$
- D. $2 - \frac{3\pi}{4}$
-
- 4** Which of the following is equivalent to $\left| \text{proj}_{\underline{v}} \underline{u} \right|$? **1**
- A. $|\underline{u}|$
- B. $\frac{\underline{u} \cdot \underline{v}}{|\underline{v}|}$
- C. $\frac{|\underline{u} \cdot \underline{v}|}{|\underline{v}|}$
- D. $\frac{\underline{u} \cdot \underline{v}}{|\underline{u}|}$
-
- 5** Which of the following gives the same curve as $x = \sin t$ and $y = -t$, for all real numbers t ? **1**
- A. $x = \sin(2t), y = 2t$
- B. $x = -\sin(2t + \pi), y = 2t + \pi$
- C. $x = -\sin(t^2 + \pi), y = t^2 + \pi$
- D. $x = \sin(t^2), y = -t^2$

- 6** How many seven letter words can be formed from the word THUNDER, such that T and R must be at either end, and U is somewhere to the right of D? **1**

A. 60
B. 120
C. 240
D. None of the above.

- 7** What is the number of solutions to the equation $x + e^{-x} = \left||x| - 2\right|$? **1**

A. 0
B. 1
C. 2
D. 3

- 8** The gradient of the normal at any point P on a particular curve is half the gradient of the line joining P and the origin. **1**

Which of the following differential equations does this curve satisfy?

A. $\frac{dy}{dx} + \frac{x}{2y} = 0$
B. $\frac{dy}{dx} - \frac{x}{2y} = 0$
C. $\frac{dy}{dx} + \frac{2x}{y} = 0$
D. $\frac{dy}{dx} - \frac{2x}{y} = 0$

- 9** Let $f(x) = \sin(2x) - \sqrt{3} \cos(2x)$. **1**

What is the value of $\int_0^{\frac{\pi}{2}} \sin^{-1} \left(\frac{f(x)}{2} \right) dx$?

- A. $\frac{\pi^2}{12}$
- B. $\frac{\pi^2}{18}$
- C. $\frac{5\pi^2}{36}$
- D. $\frac{5\pi^2}{72}$

- 10** Distinct factors of 337 500 are randomly selected. **1**

What is the minimum number of distinct factors of 337 500 that must be selected to guarantee that the product of these selected factors will be a multiple of 337 500?

- A. 22
- B. 26
- C. 31
- D. 37

Examination continues overleaf...

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

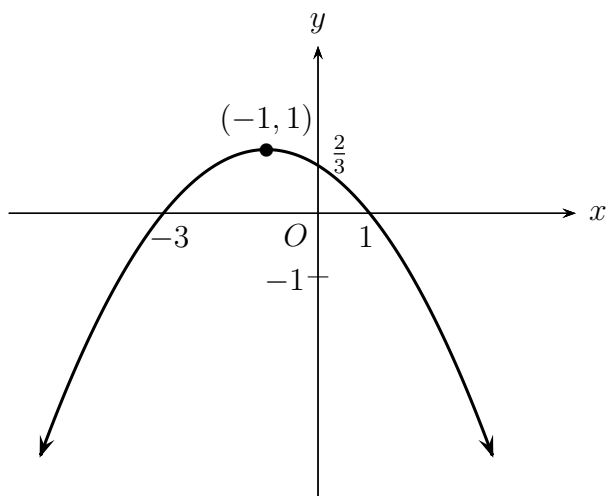
Answer each question on the writing paper supplied. Start each question on a NEW page. Extra writing paper are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start on a NEW page		Marks
(a)	Differentiate $(\tan^{-1} x)^2$.	2
(b)	Use the substitution $u = x + 2$ to find $\int (x - 1)\sqrt{x + 2} \, dx$.	3
(c)	Solve $\frac{x}{1 - x^2} < 0$.	3
(d)	Consider the function $f(x) = \frac{x - 2}{x - 5}$.	
	(i) State the range of $f^{-1}(x)$.	1
	(ii) Find an expression for $f^{-1}(x)$.	2

Examination continues overleaf...

- (e) Consider the graph of the function $y = f(x)$ shown below (which is also provided on page 6 of the writing booklet.)



On the diagrams provided in the writing booklet, sketch the graphs of the following, clearly showing all important features.

(i) $y = \frac{1}{f(x)}$ **2**

(ii) $y^2 = |f(x)|$ **2**

End of Question 11

Question 12 (15 marks) Start on a NEW page**Marks**

- (a) Find $\int 3 \cos(2x) \cos\left(5x - \frac{\pi}{3}\right) dx$. **2**
- (b) Solve $2 \operatorname{cosec} \theta - 4 \cot \theta = 3$ using the t -formula, for $0 \leq \theta \leq 2\pi$. Give your answer(s) correct to 3 significant figures. **3**
- (c) The region enclosed by $y = e^{-x}$, $y = x$, $x = 1$ and $x = 3$ is rotated about the x -axis. **3**
- Calculate the volume of the solid of revolution. Give your answer in fully simplified exact form.
- (d) Let $\underline{u} = 2\underline{i} + 5\underline{j}$ and $\underline{v} = -3\underline{i} + \lambda^2\underline{j}$, where λ is a real number.
- (i) Determine the values of λ such that $\underline{u}$ and $\underline{v}$ are perpendicular. **2**
- (ii) Explain why $\underline{u}$ cannot be parallel to $\underline{v}$. (Justify your answer mathematically.) **2**
- (iii) If $\lambda = 1$, determine $\underline{q}$ such that $\underline{u} = \operatorname{proj}_{\underline{v}}(\underline{u}) + \underline{q}$. **3**

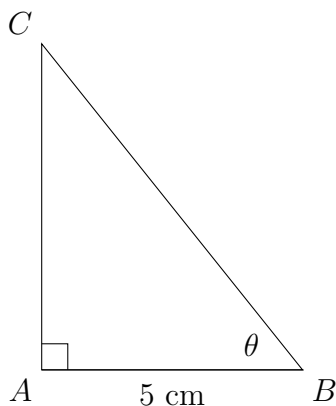
End of Question 12

Question 13 (15 marks) Start on a NEW page

Marks

- (a) The roots of $P(x) = 8x^3 - 5x^2 + 2x - 1$ are α , β and γ . **2**
Find the value of $\alpha^2 + \beta^2 + \gamma^2$.

- (b) The triangle ABC is right angled at A as shown in the diagram. **3**
Let $AB = 5$ cm and $\angle ABC = \theta$ radians.



If θ is increasing at a constant rate of 0.1 radians per second, find the rate at which the perimeter of the triangle is increasing when $\theta = \frac{\pi}{3}$ radians. Leave your answer in exact form.

- (c) Prove by mathematical induction that, for all integers $n \geq 1$, **3**

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \cdots + (n^2 + 1) \times n! = n \times (n + 1)!$$

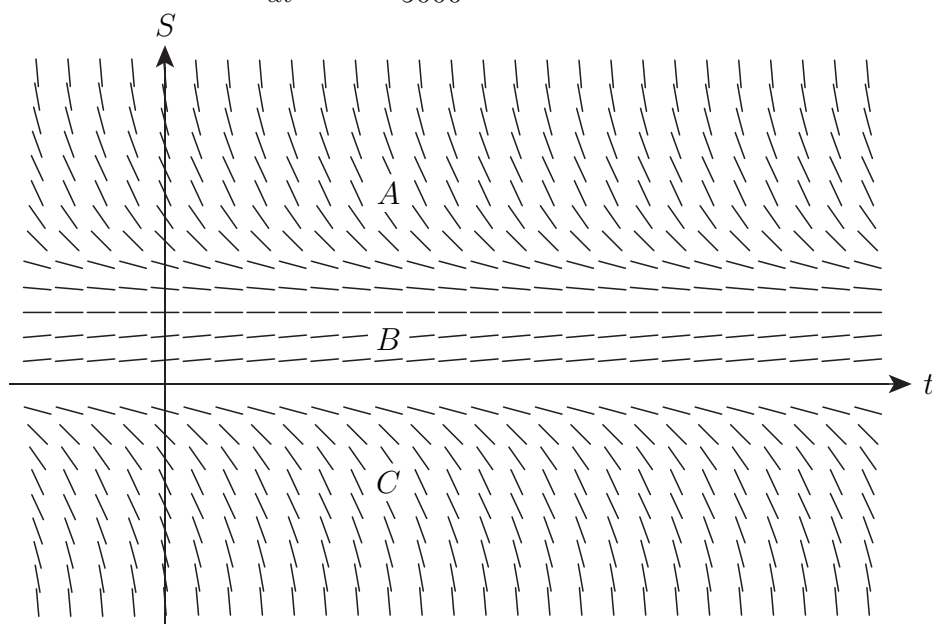
Examination continues overleaf...

- (d) The rate of spread of a particular flu through a population is modelled by the equation

$$\frac{dS}{dt} = \frac{S(2000 - S)}{5000},$$

where S is the total number of people infected after t days.

- (i) Show that $\frac{1}{2000} \left(\frac{1}{S} + \frac{1}{2000 - S} \right) = \frac{1}{S(2000 - S)}$. **1**
- (ii) Solve the above differential equation using separation of variables to show that $S = \frac{2000}{1 + Qe^{-0.4t}}$, where Q is a constant. **3**
- (iii) The slope field of $\frac{dS}{dt} = \frac{S(2000 - S)}{5000}$ is shown below. **1**



Consider the particular solution $S = \frac{2000}{1 + 199e^{-0.4t}}$.

There are three regions labelled A , B and C .

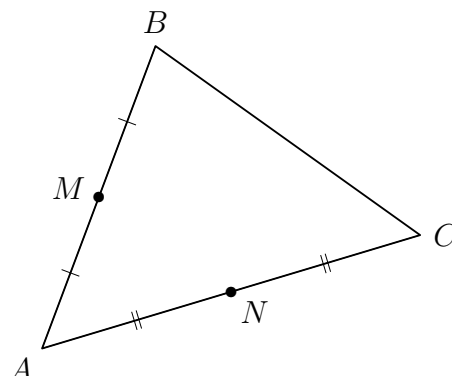
In which one of these three regions will the curve of the particular solution lie in? You must justify your answer.

- (iv) According to this model, and using $Q = 199$, after how many days does the rate of spread reach its maximum? Give your answer correct to 2 decimal places. **2**

End of Question 13

Question 14 (15 marks) Start on a NEW page**Marks**

- (a) In $\triangle ABC$ below, M and N are the midpoints of AB and AC respectively.



Let $\overrightarrow{AB} = \mathbf{b}$ and $\overrightarrow{AC} = \mathbf{c}$.

- (i) Show that $\overrightarrow{BN} = \frac{1}{2}\mathbf{c} - \mathbf{b}$, and write a similar expression for $\overrightarrow{CM}$. **2**

- (ii) Hence, using vectors, show that if BN and CM are perpendicular, then $|\mathbf{b}|^2 + |\mathbf{c}|^2 = 5|\mathbf{a}|^2$, where $\mathbf{a} = \overrightarrow{BC}$. **3**

- (b) The position vectors of particles P and Q at time t minutes are

$$\begin{pmatrix} \cos t \\ \sin t \end{pmatrix} \text{ and } \begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix}$$

respectively, where $0 \leq t \leq 2\pi$.

Let θ be the angle between the vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ at time t , where O is the origin.

- (i) Show that $\cos \theta = \cos 2t$. **2**

- (ii) Find the total time for which $\theta \geq \frac{\pi}{4}$. **2**

Examination continues overleaf...

- (c) A particle is projected from the origin O with initial speed U and angle of elevation α , where $0 < \alpha < \frac{\pi}{2}$.

The position vector of the particle, $\underline{r}(t)$, where t is the time after projection and g is the acceleration due to gravity, is given by

$$\underline{r}(t) = \begin{pmatrix} Ut \cos \alpha \\ Ut \sin \alpha - \frac{gt^2}{2} \end{pmatrix}. \quad (\text{Do NOT prove this.})$$

- (i) Prove that if the particle passes through the point $P(a, b)$, where $a > 0$, then the minimum possible value of U is **3**

$$\sqrt{g \left(b + \sqrt{a^2 + b^2} \right)}.$$

Suppose that for parts (ii) and (iii), the particle is projected with the minimum possible speed in order to pass through the point $P(a, b)$.

- (ii) Show that $\tan \alpha = \frac{b + \sqrt{a^2 + b^2}}{a}$. **1**
- (iii) Deduce that $\underline{v}(0)$ bisects the angle between OP and the positive y -axis, where $\underline{v}(0)$ is the initial velocity vector of the particle. **2**

End of paper



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10 marks

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Questions

Marks

1 What is the magnitude of $2\underline{u} + 3\underline{v}$?

1

A. $\sqrt{5}$

B. $\sqrt{11}$

C. $\sqrt{13}$

☒ D. Cannot be determined.

$\underline{u}, \underline{v}$ are not known.

$\therefore \textcircled{D}$

2 The polynomial $P(x) = x^3 + 5x^2 + bx + c$ has a double root at $x = -1$.

1

What are the values of b and c ?

☒ A. $b = 7, c = 3$

B. $b = 7, c = -3$

C. $b = -7, c = 3$

D. $b = -7, c = -3$

$$P'(x) = 3x^2 + 10x + b$$

$$\therefore 3(-1)^2 + 10(-1) + b = 0$$

$$-7 + b = 0$$

$$\therefore b = 7$$

$$(-1)^3 + 5(-1)^2 + 7(-1) + c = 0$$

$$-3 + c = 0$$

$$\therefore c = 3$$

- 3 What is the value of $\int_0^{\frac{\pi}{4}} (4 \cos^2 x - 5) dx$? 1

A. $1 - \frac{3\pi}{4}$

B. $1 - \frac{5\pi}{4}$

C. $-1 - \frac{3\pi}{4}$

D. $2 - \frac{3\pi}{4}$

$$= \int_0^{\frac{\pi}{4}} 2(\cos 2x + 1) - 5 dx$$

$$= \int_0^{\frac{\pi}{4}} 2 \cos 2x - 3 dx$$

$$= \left[\sin 2x - 3x \right]_0^{\frac{\pi}{4}}$$

$$= 1 - \frac{3\pi}{4}$$

- 4 Which of the following is equivalent to $|\text{proj}_{\underline{v}} \underline{u}|$? 1

A. $|\underline{u}|$

B. $\frac{\underline{u} \cdot \underline{v}}{|\underline{v}|}$

C. $\frac{|\underline{u} \cdot \underline{v}|}{|\underline{v}|}$

D. $\frac{\underline{u} \cdot \underline{v}}{|\underline{u}|}$

$$\left| \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \underline{v} \right|$$

$$= \frac{|\underline{u} \cdot \underline{v}|}{|\underline{v}|^2} |\underline{v}|$$

$$= \text{C}$$

- 5 Which of the following gives the same curve as $x = \sin t$ and $y = -t$, for all real numbers t ? 1

A. $x = \sin(2t), y = 2t$

B. $x = -\sin(2t + \pi), y = 2t + \pi$

C. $x = -\sin(t^2 + \pi), y = t^2 + \pi$

D. $x = \sin(t^2), y = -t^2$

$\rightarrow x = \sin y$ X

$x = \sin(-y)$
 $x = -\sin y$

] Not these since y needs to be able to attain all real values.

- 6 How many seven letter words can be formed from the word THUNDER, such that T and R must be at either end, and U is somewhere to the right of D? 1

A. 60

☒ B. 120

C. 240

D. None of the above.

$$T \left(\frac{5!}{2!} \right) R$$

$$\frac{5!}{2!} \times 2! = 120$$

swap T and R

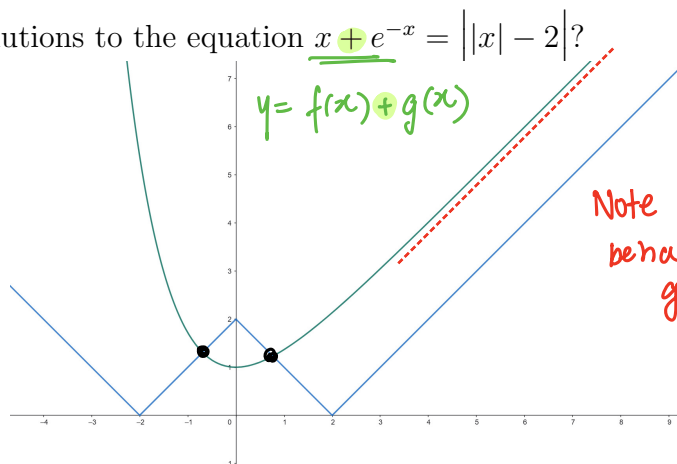
- 7 What is the number of solutions to the equation $x + e^{-x} = ||x| - 2|$? 1

A. 0

B. 1

☒ C. 2

D. 3



- 8 The gradient of the normal at any point P on a particular curve is half the gradient of the line joining P and the origin. 1

Which of the following differential equations does this curve satisfy?

A. $\frac{dy}{dx} + \frac{x}{2y} = 0$

B. $\frac{dy}{dx} - \frac{x}{2y} = 0$

☒ C. $\frac{dy}{dx} + \frac{2x}{y} = 0$

D. $\frac{dy}{dx} - \frac{2x}{y} = 0$

$$-\frac{1}{m_T} = \frac{1}{2} m_{OP}$$

$$-\frac{1}{\frac{dy}{dx}} = \frac{1}{2} \frac{y}{x}$$

$$\therefore -\frac{dy}{dx} = \frac{2x}{y}$$

☒ C

flip.

- 9 Let $f(x) = \sin(2x) - \sqrt{3}\cos(2x)$.

$$f(x) = 2 \sin\left(2x - \frac{\pi}{3}\right) \quad (\text{Auxiliary } \angle)$$

What is the value of $\int_0^{\frac{\pi}{2}} \sin^{-1}\left(\frac{f(x)}{2}\right) dx$?

A. $\frac{\pi^2}{12}$

B. $\frac{\pi^2}{18}$

C. $\frac{5\pi^2}{36}$

D. $\frac{5\pi^2}{72}$

$$\begin{aligned}
 &= \int_0^{\frac{\pi}{2}} \sin^{-1}\left(\sin\left(2x - \frac{\pi}{3}\right)\right) dx \quad u = 2x - \frac{\pi}{3} \\
 &\quad du = 2 dx \\
 &= \frac{1}{2} \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sin^{-1}(\sin u) du \\
 &= \frac{1}{2} \times A \\
 &= \frac{1}{2} \times \frac{\pi}{6} \left(\frac{\pi}{3} + \frac{\pi}{2}\right) \times 2 \\
 &= \frac{\pi}{12} \left(\frac{5\pi}{6}\right) = \text{D}
 \end{aligned}$$

- 10 Distinct factors of 337 500 are randomly selected.

1

What is the minimum number of distinct factors of 337 500 that must be selected to guarantee that the product of these selected factors will be a multiple of 337 500?

A. 22

B. 26

C. 31

D. 37

$$337500 = 2^2 \times 3^3 \times 5^5$$

WORST CASE SCENARIO :

- select all factors with no 2's
i.e. factors of the form

$$3^a \times 5^b \quad \therefore 4 \times 6 = \underline{24} \text{ factors}$$

- select a factor with 2^1

$\therefore$ product will be $2 \times 3^a \times 5^b$ ($a \geq 3, b \geq 5$)
i.e. still NOT a multiple of 337 500

- select another factor [which will contain 2, since factors with no 2's are all chosen]

Examination continues overleaf...

$\therefore$ product will be

$$2^c \times 3^a \times 5^b \quad (c \geq 2, a \geq 3, b \geq 5)$$

i.e. it is a MULTIPLE of 337 500

$$\therefore 24 + 1 + 1 = 26 \text{ factors needed.}$$

Question 11

$$\begin{aligned} (a) \quad y' &= 2(\tan^{-1}x)' \times \frac{d}{dx}(\tan^{-1}x) \\ &= \frac{2 \tan^{-1}x}{1+x^2} \quad \checkmark \checkmark \end{aligned}$$

Marker's Comments

- Generally well done

$$\begin{aligned} (b) \quad &\int (x-1)\sqrt{x+2} \, dx \quad \begin{array}{l} u=x+2 \\ du=dx, \quad x=u-2 \end{array} \\ &= \int (u-2-1)\sqrt{u} \, du \quad \checkmark \end{aligned}$$

$$= \int u^{\frac{3}{2}} - 3u^{\frac{1}{2}} \, du$$

$$= \frac{2u^{\frac{5}{2}}}{5} - \frac{3u^{\frac{3}{2} \times 2}}{3} + C \quad \checkmark$$

$$= \frac{2}{5}(x+2)^{\frac{5}{2}} - 2(x+2)^{\frac{3}{2}} + C \quad \checkmark$$

Marker's Comments

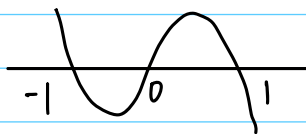
- Generally well done. The most common error was not replacing u in terms of x .

$$(c) \quad \frac{x}{1-x^2} < 0$$

$$\frac{x(1-x^2)^2}{1-x^2} < 0 \times (1-x^2)^2$$

$$x(1-x^2) < 0 \quad \checkmark$$

$$x(1+x)(1-x) < 0$$



$$\therefore -1 < x < 0, x > 1 \quad \checkmark \checkmark$$

Marker's Comments

- Generally well done. Most common errors included:

- writing $x(1-x^2) < (1-x)^2$
- writing $x(1-x^2)^2 < 0$
- not sketching correctly
- inclusion of either $x = -1, 0, 1$

(d)

(i) Domain of $f(x)$: All real x , $x \neq 5$

$\therefore$ Range of $f^{-1}(x)$: All real y , $y \neq 5$. ✓

(ii) $x = \frac{y-2}{y-5}$ ✓

$$xy - 5x = y - 2$$

$$y(x-1) = 5x-2$$

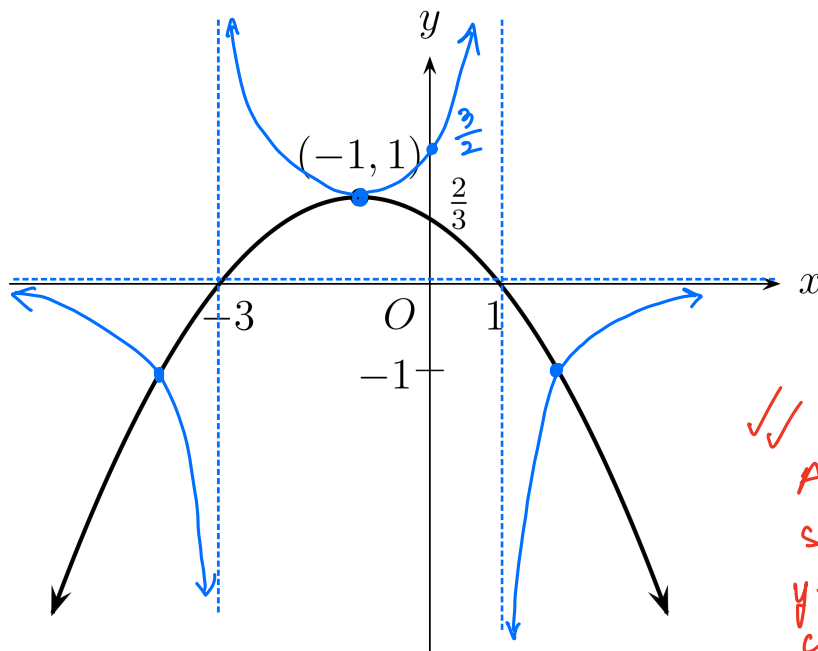
$$\therefore f^{-1}(x) = \frac{5x-2}{x-1}$$
 ✓

Marker's Comments

- Generally well done

(e)

(i)

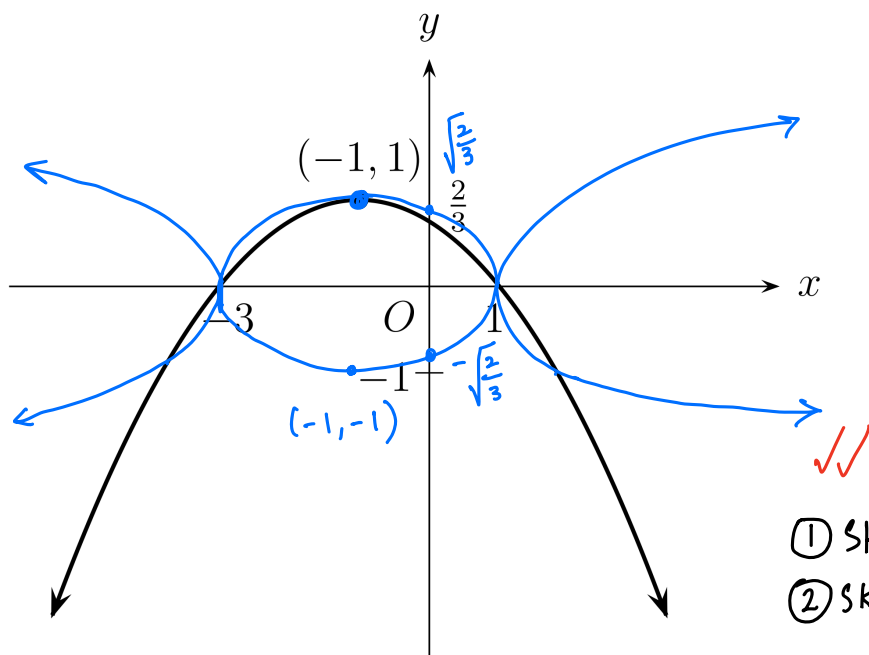


✓✓

All features must be shown, including y-intercepts and common points when $y = 1$ and $y = -1$

(e)

(ii)



Marker's Comments

- Generally not well done

- Many students did not include the y-intercepts, or the reflection across the x-axis.
- Some also had their graph below the original for $-3 \leq x \leq 1$.

Question 12(a)	Marks
· Provides correct solution.	2 marks
· Applies appropriate product to sum formula correctly.	1 mark

Solution

From the reference sheet we have the following product to sum formula:

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)].$$

Let $A = 2x$ and $B = 5x - \frac{\pi}{3}$:

$$\begin{aligned} \cos(2x) \cos\left(5x - \frac{\pi}{3}\right) &= \frac{1}{2} \left[\cos\left(2x - \left(5x - \frac{\pi}{3}\right)\right) + \cos\left(2x + \left(5x - \frac{\pi}{3}\right)\right) \right] \\ &= \frac{1}{2} \left[\cos\left(\frac{\pi}{3} - 3x\right) + \cos\left(7x - \frac{\pi}{3}\right) \right] \checkmark \\ &= \frac{1}{2} \left[\cos\left(3x - \frac{\pi}{3}\right) + \cos\left(7x - \frac{\pi}{3}\right) \right] \end{aligned}$$

We used the fact that cosine is even. Therefore

$$\begin{aligned} \int 3 \cos(2x) \cos\left(5x - \frac{\pi}{3}\right) dx &= \frac{3}{2} \int \left(\cos\left(3x - \frac{\pi}{3}\right) + \cos\left(7x - \frac{\pi}{3}\right) \right) dx \\ &= \frac{3}{2} \left(\frac{1}{3} \sin\left(3x - \frac{\pi}{3}\right) + \frac{1}{7} \sin\left(7x - \frac{\pi}{3}\right) \right) + C \\ &= \frac{1}{2} \sin\left(3x - \frac{\pi}{3}\right) + \frac{3}{14} \sin\left(7x - \frac{\pi}{3}\right) + C \checkmark \end{aligned}$$

Comments

- Some students used the angle expansion formula from the beginning, which needlessly complicated the expression.
- Some students did not realise that $\cos(7x - \frac{\pi}{3})$ can be integrated directly, and instead used the angle expansion formula.

Question 12(b)	Marks
· Provides correct solution.	3 marks
· Obtains $6t^2 - 6t - 2 = 0$.	2 marks
· Substitutes relevant t -results.	1 mark

Solution

Let $t = \tan \frac{\theta}{2}$.

$$\frac{2}{\sin \theta} - \frac{4}{\tan \theta} = 3$$

$$\frac{2}{\frac{2t}{1+t^2}} - \frac{4}{\frac{2t}{1-t^2}} = 3$$

$$2(1+t^2) - 4(1-t^2) = 6t \quad \checkmark$$

$$2 + 2t^2 - 4 + 4t^2 = 6t$$

$$6t^2 - 6t - 2 = 0$$

$$3t^2 - 3t - 1 = 0$$

$$t = \frac{3 \pm \sqrt{21}}{6} \quad \checkmark$$

$$\tan \frac{\theta}{2} = \frac{3 \pm \sqrt{21}}{6}$$

$$\frac{\theta}{2} = \arctan \left(\frac{3 + \sqrt{21}}{6} \right), \quad \pi + \arctan \left(\frac{3 - \sqrt{21}}{6} \right)$$

$$\theta = 1.80, 5.77 \text{ (3 sig. fig.)} \quad \checkmark$$

Note that there is no need to check the solution $\theta = \pi$, since it leads to division by zero in the original expression.

Comments

- Most students were able to find the first solution of $\theta = 1.80$, but many were unable to find the second solution $\theta = 5.77$.
- Many students wrote negative values of θ , which were not in the domain $0 \leq \theta \leq 2\pi$.
- Students are strongly encouraged to revise solving trigonometric equations.

Question 12(c)	Marks
· Provides correct solution.	3 marks
· Carries through at least one of the resulting integrals correctly.	2 marks
· Sets up top volume minus bottom volume as shown below.	1 mark

Solution

Use the principle of “top volume minus bottom volume”.

$$\begin{aligned}
 V &= \pi \int_1^3 y_{\text{top}}^2 dx - \pi \int_1^3 y_{\text{bottom}}^2 dx \quad \checkmark \\
 &= \pi \int_1^3 (x^2 - e^{-2x}) dx \\
 &= \pi \left[\frac{1}{3}x^3 + \frac{1}{2}e^{-2x} \right]_1^3 \quad \checkmark \\
 &= \pi \left(\frac{26}{3} + \frac{1}{2}e^{-6} - \frac{1}{2}e^{-2} \right) \quad \checkmark
 \end{aligned}$$

Comments

- 1 mark was awarded for either finding $\pi \int_1^3 x^2 dx = \frac{26\pi}{3}$ or $\pi \int_1^3 e^{-2x} dx = \pi (e^{-2} - e^{-6})$.
- After a fully simplified correct answer was found, subsequent simplification errors were ignored.

• There were common mistakes in setting up the desired volume, such as $\int_1^3 (x - e^{-2x})^2 dx$ or having the top and bottom curves swapped around, that is, $\int_1^3 (e^{-2x} - x^2) dx$.

- There were occasionally some problems with the limits of integration, and some students attempted to find points of intersection (not needed).

Question 12(d)(i)	Marks
· Provides correct solution.	2 marks
· Writes $\underline{u} \cdot \underline{v} = 0$.	1 mark

Solution

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ \lambda^2 \end{bmatrix} = 0 \quad \checkmark$$

$$-6 + 5\lambda^2 = 0$$

$$\lambda^2 = \frac{6}{5}$$

$$\lambda = \pm \sqrt{\frac{6}{5}} \quad \checkmark$$

Comments

- Well done. Some students forgot the plus/minus sign, and others had the fraction upsidedown: $\lambda = \pm \sqrt{\frac{5}{6}}$.

Question 12(d)(ii)	Marks
· Provides correct solution.	2 marks
· Introduces a scalar of proportionality and solves simultaneously.	1 mark

Solution

If $\underline{u}$ and $\underline{v}$ were parallel, there would exist a real scalar r such that $\underline{u} = r\underline{v}$.(*)

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -3r \\ \lambda^2 r \end{bmatrix}$$

$$2 = -3r \quad (1)$$

$$5 = \lambda^2 r \quad (2) \quad \checkmark$$

From (1) we have $r = -\frac{2}{3}$.

Substitute this into (2) to get $5 = \lambda^2 \times -\frac{2}{3}$, that is, $\lambda^2 = 5 \times -\frac{3}{2} = -7.5$.

This is impossible, because the square of a real number can't be negative.

Therefore no such r exists. Thus $\underline{u}$ and $\underline{v}$ cannot be parallel. $\checkmark$

Comments

- Students often incorrectly assumed that $\lambda^2 = \frac{6}{5}$, but this does not apply in (ii), it only applies for perpendicular lines, as found in (i).
- The scalar multiple criterion for two vectors to be parallel labeled (*) above was not well understood and should be revised by students.
- Some students gave the following elegant geometrical argument: $\underline{u} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ lives in the first quadrant, but because $\lambda^2 \geq 0$, $\underline{v}$ lives in the second quadrant. Therefore they have no chance of being parallel.
- Some students successfully avoided explicitly introducing a scalar by instead considering a ratio of components.

Question 12(d)(iii)	Marks
· Provides correct solution.	3 marks
· Calculates relevant dot products correctly.	2 marks
· States projection formula correctly.	1 mark

Solution

$$\underline{u} = \text{proj}_{\underline{v}} \underline{u} + \underline{q}$$

$$\underline{q} = \underline{u} - \text{proj}_{\underline{v}} \underline{u}$$

$$= \begin{bmatrix} 2 \\ 5 \end{bmatrix} - \frac{\underline{u} \cdot \underline{v}}{\underline{v} \cdot \underline{v}} \underline{v} \quad \checkmark$$

$$= \begin{bmatrix} 2 \\ 5 \end{bmatrix} - \frac{-1}{10} \begin{bmatrix} -3 \\ 1 \end{bmatrix} \quad \checkmark$$

$$= \begin{bmatrix} 2 - 0.3 \\ 5 + 0.1 \end{bmatrix}$$

$$= \begin{bmatrix} 1.7 \\ 5.1 \end{bmatrix} \quad \checkmark$$

Comments

- Quite well done. A common mistake was to forget to square $|\underline{v}|$ in the denominator of the projection formula. It is safer to use the formula shown above with $\underline{v} \cdot \underline{v}$ in the denominator.
- Another common mistake was to forget to multiply $\underline{q}$ by 10 when multiplying both sides by 10 on this line: $\begin{bmatrix} 2 \\ 5 \end{bmatrix} = -\frac{1}{10} \begin{bmatrix} -3 \\ 1 \end{bmatrix} + \underline{q}$.
- Students commonly swapped $\underline{u}$ and $\underline{v}$ by mistake in the projection formula. If this was carried out successfully, the correct answer would be $\frac{30}{29} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

Q13 a) $p(x) = 8x^3 - 5x^2 + 2x - 1$

$$\alpha + \beta + \gamma = -\frac{b}{a} = \frac{5}{8}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{1}{4}$$

$$\alpha\beta\gamma = -\frac{d}{a} = \frac{1}{8}$$

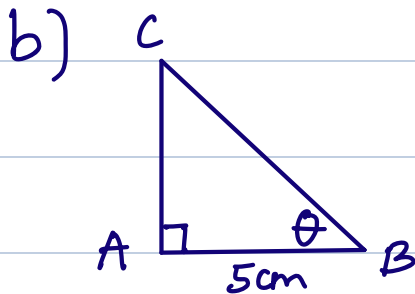
$$(\alpha + \beta + \gamma)(\alpha + \beta + \gamma) = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$\alpha^2 + \beta^2 + \gamma^2 = \left(\frac{5}{8}\right)^2 - 2\left(\frac{1}{4}\right) = -\frac{7}{64}$$

• Expand $(\alpha + \beta + \gamma)^2$ correctly : 1M

• Correct working and solution : 2M



$$\frac{d\theta}{dt} = 0.1 \text{ rad/s}$$

Find $\frac{dp}{dt}$ when $\theta = \frac{\pi}{3}$

$$\tan \theta = \frac{AC}{5} \rightarrow AC = 5 \tan \theta$$

$$\cos \theta = \frac{5}{BC} \rightarrow BC = \frac{5}{\cos \theta}$$

$$p = 5 + 5 \tan \theta + \frac{5}{\cos \theta}$$

$$\frac{dp}{d\theta} = 5 \sec^2 \theta + \frac{5 \sin \theta}{\cos^2 \theta} = 5 \sec^2 \theta + 5 \tan \theta \sec \theta$$

$$\frac{dp}{dt} = \frac{dp}{d\theta} \times \frac{d\theta}{dt} = \frac{1}{10} (5 \sec^2 \theta + 5 \tan \theta \sec \theta)$$

when $\theta = \frac{\pi}{3}$

$$\frac{dp}{dt} = \frac{1}{2} \left(\sec^2 \frac{\pi}{3} + \tan \frac{\pi}{3} \cdot \sec \frac{\pi}{3} \right)$$

$$= \frac{1}{2} \left(\frac{1}{1/4} + \sqrt{3} \times \frac{1}{1/2} \right)$$

$$= \frac{1}{2} (4 + 2\sqrt{3}) = (2 + \sqrt{3}) \text{ cm/s} \checkmark$$

• correct expression $P = 5 + 5 \tan \theta + \frac{5}{\cos \theta} : 1M$

• correct $\frac{dp}{dt}$ in terms of $\theta : 2M$

• correct working and solution : 3M

C) prove true for $n=1$

$$\text{LHS} = (1^2 + 1) \times 1! = 2 \checkmark$$

$$\text{RHS} = 1 \times 2! = 2 \quad \therefore \text{LHS} = \text{RHS}$$

Assume it is true for $n=k$

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k! = k(k+1)!$$

Prove it is true for $n=k+1$

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k! + \\ \left[(k+1)^2 + 1 \right] \times (k+1)! = (k+1)(k+2)!$$

$$k(k+1)! + [(k+1)^2 + 1] \times (k+1)! = (k+1)(k+2)! \quad \checkmark$$

from assumption

$$\begin{aligned} \text{LHS} &= (k+1)! (k + k^2 + 2k + 1 + 1) \\ &= (k+1)! (k^2 + 3k + 2) \\ &= (k+1)! (k+2)(k+1) \\ &= (k+1)(k+2)! \quad \checkmark \end{aligned}$$

$$= \text{RHS} \quad \text{True for } n = k+1$$

$\therefore$ proven by mathematical induction.

- prove it is true for $n=1$: 1M
- Subs the assumption for step $n=k+1$: 2M
- Correct prove : 3M

d)

$$\begin{aligned} \text{i) LHS} &= \frac{1}{2000} \left[\frac{2000 - \cancel{S} + \cancel{S}}{S(2000 - S)} \right] \quad \checkmark \\ &= \frac{1}{S(2000 - S)} \end{aligned}$$

$$\text{ii) } \frac{ds}{dt} = \frac{S(2000 - S)}{5000}$$

$$\int \frac{1}{S(2000 - S)} ds = \frac{1}{5000} \int dt$$

$$\frac{1}{2000} \int \left(\frac{1}{s} + \frac{1}{2000-s} \right) ds = \frac{t}{5000} + C$$

$$\frac{1}{2000} \left[\ln|s| - \ln|2000-s| \right] = \frac{t}{5000} + C \quad \checkmark$$

$$\ln \left| \frac{s}{2000-s} \right| = 0.4t + C$$

$$\frac{s}{2000-s} = e^C \cdot e^{0.4t} \quad \checkmark$$

$$\frac{s}{2000-s} = A e^{0.4t} \quad (e^C = A)$$

$$s = A e^{0.4t} (2000 - s)$$

$$s + s \cdot A e^{0.4t} = 2000 \cdot A e^{0.4t}$$

$$s(1 + A e^{0.4t}) = 2000 A e^{0.4t}$$

$$s = \frac{2000 A e^{0.4t}}{1 + A e^{0.4t}}$$

$$s = \frac{2000}{1 + A e^{-0.4t}} \quad \checkmark \quad \text{Divide by } A e^{0.4t} \quad (A = \frac{1}{A})$$

Marks are awarded for each stage $\checkmark$ above.

13/d/iii)

$$t \rightarrow \infty, S \rightarrow 2000$$

$$t = 0, S = \frac{2000}{200} = 10$$

In region A, $S > 2000$, thus Not A

In region C, $S \leq 0$ Not C

$S = 10$ lies in region B

$\therefore$ Region B ✓

13/d/iv)

$\frac{ds}{dt}$ is maximum if $\frac{d^2s}{dt^2} = 0$,
the point where the gradient changes
from increasing to decreasing.

For logistic model $S = \frac{A}{1 + Be^{-kt}}$

Max growth occurs when $S = \frac{A}{2}$

$$S = \frac{2000}{1 + 199e^{-0.4t}} \therefore S = \frac{2000}{2} = 1000 \quad \checkmark$$

$$1000 = \frac{2000}{1 + 199e^{-0.4t}} \therefore 1 + 199e^{-0.4t} = 2$$

$$e^{-0.4t} = \frac{1}{199} \rightarrow -0.4t = \ln \frac{1}{199}$$

$$t = 13.23 \text{ days} \quad \checkmark$$

Recognise Max growth occurs when $S = 1000$: 1M

Differentiate ds/dt correctly: 1M

Question 14

(a) (i) $\vec{BN} = \vec{BA} + \vec{AN}$ $\vec{AB} = \vec{b}$ $\vec{AC} = \vec{c}$

$$\begin{aligned} &= -\vec{AB} + \frac{1}{2}\vec{AC} \\ &= -\vec{b} + \frac{1}{2}\vec{c} \\ &= \frac{1}{2}\vec{c} - \vec{b}, \text{ as required} \end{aligned}$$

✓ must be clearly proven.

$$\begin{aligned} \vec{CM} &= \vec{CA} + \vec{AM} \\ &= -\vec{c} + \frac{1}{2}\vec{b} \\ &= \frac{1}{2}\vec{b} - \vec{c} \end{aligned}$$

✓

(ii) If perpendicular $\vec{BN} \cdot \vec{CM} = 0$ prove $|\vec{b}|^2 + |\vec{c}|^2 = 5|\vec{a}|^2$

$$\therefore \left(\frac{1}{2}\vec{c} - \vec{b}\right) \cdot \left(\frac{1}{2}\vec{b} - \vec{c}\right) = 0$$

$$\frac{1}{4}\vec{b} \cdot \vec{c} - \frac{1}{2}\vec{c} \cdot \vec{c} - \frac{1}{2}\vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{b} = 0$$

$$\frac{5}{4}\vec{b} \cdot \vec{c} - \frac{1}{2}|\vec{c}|^2 - \frac{1}{2}|\vec{b}|^2 = 0$$

$$\begin{aligned} \frac{5}{4}\vec{b} \cdot \vec{c} &= \frac{1}{2}(|\vec{b}|^2 + |\vec{c}|^2) \\ \vec{b} \cdot \vec{c} &= \frac{2}{5}(|\vec{b}|^2 + |\vec{c}|^2) \end{aligned}$$

✓ 1 mark (1)

$$\begin{aligned} \vec{a} &= \vec{BC} \\ &= -\vec{b} + \vec{c} \\ &= \vec{c} - \vec{b} \end{aligned}$$

Note: Alternatively, apply cosine rule.

$$|\vec{a}|^2 = |\vec{c} - \vec{b}|^2$$

$$|\vec{a}|^2 = |\vec{b}|^2 + |\vec{c}|^2 - 2\vec{b} \cdot \vec{c} \quad (2)$$

✓ 2nd mark
sub (1) in (2)

$$|\vec{a}|^2 = |\vec{b}|^2 + |\vec{c}|^2 - 2 \times \frac{2}{5}(|\vec{b}|^2 + |\vec{c}|^2)$$

$$5|\vec{a}|^2 = 5|\vec{b}|^2 + 5|\vec{c}|^2 - 4|\vec{b}|^2 - 4|\vec{c}|^2$$

$$\therefore 5|\vec{a}|^2 = |\vec{b}|^2 + |\vec{c}|^2, \text{ as required}$$

✓ 3rd mark
for correct substitution.

$$(b) (i) \vec{OP} = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} \quad \vec{OQ} = \begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix}$$

$$\cos Q = \frac{\vec{OP} \cdot \vec{OQ}}{|\vec{OP}| |\vec{OQ}|}$$

$$\cos Q = \frac{\cos t \cdot \cos 3t + \sin t \cdot \sin 3t}{\sqrt{\cos^2 t + \sin^2 t} \times \sqrt{\cos^2 3t + \sin^2 3t}}$$

$$\cos Q = \cos(3t - t)$$

$$\text{Since } \cos(A-B) = \cos A \cdot \cos B + \sin A \cdot \sin B$$

$$\therefore \cos Q = \cos 2t, \text{ as required}$$

Not necessary to use product to sum formula.

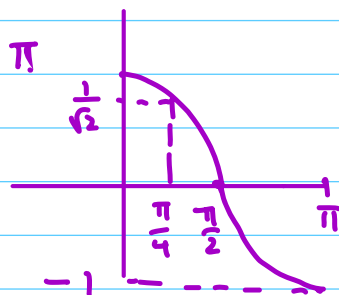
(ii) Note: angle b/w vectors $0 \leq Q \leq \pi$

In this case $Q > \pi/4 \therefore \pi/4 \leq Q \leq \pi$

$\cos Q$ is always $\searrow$ ing in this interval

$$\therefore \cos \frac{\pi}{4} > \cos Q > \cos \pi$$

$$\frac{1}{\sqrt{2}} > \cos Q > -1$$



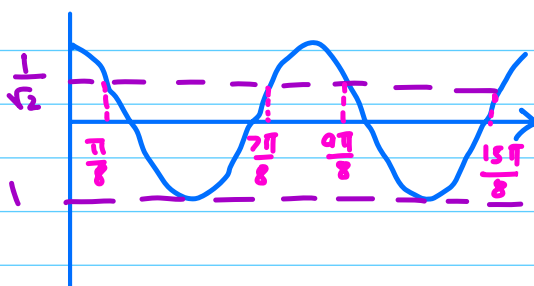
Since $\cos Q = \cos 2t$ (proven in (i))

$$\frac{1}{\sqrt{2}} > \cos 2t > -1$$

$$-1 \leq \cos 2t \leq \frac{1}{\sqrt{2}}$$

$$\cos 2t = \frac{1}{\sqrt{2}} \quad \cos 2t = -1$$

$$2t = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{15\pi}{4}$$



$$2t = \pi, 3\pi$$

$$t = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$t = \frac{\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{15\pi}{8} \quad \checkmark \text{ 1 mark for all correct solns}$$

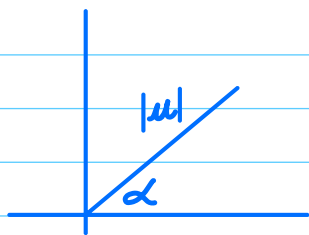
$$\therefore \text{total time for which } Q > \frac{\pi}{4} \quad \left[\frac{\pi}{8}, \frac{7\pi}{8} \right] \cup \left[\frac{9\pi}{8}, \frac{15\pi}{8} \right]$$

$$= 2 \times \frac{3\pi}{4}$$

$$= \frac{3\pi}{2} \text{ mins}$$

1 mark for correct soln with supported working.

(c)



$$x = Ut \cos d \quad y = Ut \sin d - \frac{gt^2}{2}$$

Find cartesian eqn:

$$\text{let } t = \frac{x}{U \cos d} \text{ in } y$$

$$y = \cancel{U} x \frac{x}{\cancel{U} \cos d} \sin d - \frac{g}{2} \left(\frac{x^2}{U^2 \cos^2 d} \right)$$

$$y = x \tan d - \frac{gx^2}{2U^2 \cos^2 d}$$

$$y = x \tan d - \frac{gx^2}{2U^2} \sec^2 d$$

$$y = x \tan d - \frac{gx^2}{2U^2} (\tan^2 d + 1)$$

particle passes through (a,b)

$$b = a \tan d - \frac{ga^2}{2U^2} (\tan^2 d + 1)$$

✓ 1st mark for forming correct quadratic equation.

$$\text{Let } T = \tan d$$

$$b = at - \frac{ga^2}{2U^2} (1 + T^2)$$

$$\frac{ga^2 T^2}{2U^2} - at + \left(b + \frac{ga^2}{2U^2} \right) = 0$$

Students found it challenging to progress past this point.

 $\Delta \geq 0$ for d to exist.

$$\cancel{a^2}^2 - 4 \left(\frac{ga^2}{2U^2} \right) \left(b + \frac{ga^2}{2U^2} \right) \geq 0 \quad \div a^2$$

$$1 - \frac{4g}{2U^2} \left(b + \frac{ga^2}{2U^2} \right) \geq 0$$

$$1 \geq \frac{2g}{U^2} \left(b + \frac{ga^2}{2U^2} \right)$$

$$U^2 \geq \frac{2gb + g^2 a^2}{U^2} \quad \checkmark \text{ 2nd mark}$$

$$v^4 - 2gbv^2 - g^2a^2 \geq 0$$

complete the square

$$(v^2 - gb)^2 - g^2b - g^2a^2 \geq 0$$

$$(v^2 - gb)^2 \geq g^2(a^2 + b^2)$$

$$|v^2 - gb| \geq g\sqrt{a^2 + b^2}$$

$$v^2 \geq gb + g\sqrt{a^2 + b^2}$$

$$v^2 \geq g(b + \sqrt{a^2 + b^2})$$

✓ 3rd mark

$$v \geq \sqrt{g(b + \sqrt{a^2 + b^2})}, \text{ as required.}$$

$$(ii) \quad \frac{ga^2 T^2}{2v^2} - aT + \left(b + \frac{ga^2}{2v^2}\right) = 0, \text{ where } T = \tan d$$

since v is minimum $\Delta = 0$

$$\tan d = \frac{a \pm \sqrt{0}}{2v^2}$$

$$\frac{a \pm \frac{ga^2}{2v^2}}{2v^2}$$

$$= \frac{a}{2v^2} \div \frac{ga^2}{2v^2}$$

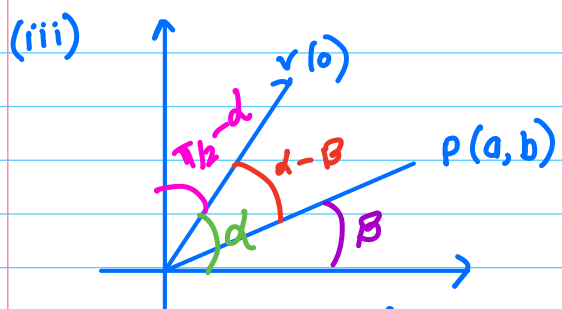
$$= \frac{v^2}{ga}$$

$$= \frac{g(b + \sqrt{a^2 + b^2})}{ga}$$

✓ 1 mark if all correct

$$= \frac{b + \sqrt{a^2 + b^2}}{a}, \text{ as required}$$

students did not know where to start with this question.



R.T.P $\frac{\pi}{2} - d = d - \beta$

ie $\tan(\frac{\pi}{2} - d) = \tan(d - \beta)$

$$\text{LHS} = \tan(\frac{\pi}{2} - d)$$

$$= \cot d$$

$$= \frac{a}{b + \sqrt{a^2 + b^2}} \quad \text{from (ii)}$$

✓ 1st mark for setting up proof and finding expression for L.H.S

$$\text{RHS} = \tan(d - \beta)$$

$$= \frac{\tan d - \tan \beta}{1 + \tan d \cdot \tan \beta}$$

$$= \frac{\frac{b + \sqrt{a^2 + b^2}}{a} - \frac{b}{a}}{1 + \left(\frac{b + \sqrt{a^2 + b^2}}{a}\right) \frac{b}{a}}$$

$$= \frac{\frac{\sqrt{a^2 + b^2}}{a}}{\frac{a^2 + b^2 + b\sqrt{a^2 + b^2}}{a^2}}$$

$$= \frac{a\sqrt{a^2 + b^2}}{a^2 + b^2 + b\sqrt{a^2 + b^2}}$$

$$= \frac{a\cancel{\sqrt{a^2 + b^2}}}{(\sqrt{a^2 + b^2})^2 + b\cancel{\sqrt{a^2 + b^2}}}$$

$$= \frac{a}{\sqrt{a^2 + b^2} + b}$$

✓ 2nd mark for correct algebra

This was a challenging question. A diagram was a helpful start.